

Use-Case Specification Report: Log Maintenance Ticket (UC-01)

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Course/Lab: Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #07: Hostel Maintenance & Issue Ticketing System

1. System Overview & Requirement Alignment

This report provides the detailed specification for Use Case UC-01 (Log Maintenance Ticket), corresponding to Functional Requirement FR-001. The objective of this use case is to provide a standardized, geo-tagged intake pipeline allowing hostel residents to log plumbing, electrical, and carpentry issues, select severity parameters, and route actionable tickets to the Maintenance Warden queue.

2. Key Use Case Metadata

Primary Actor: Hostel Resident	Supporting Actor: Maintenance Warden
Preconditions: Resident is logged into the mobile/web app with a verified room/block tag profile.	Postconditions (Success): Ticket created, assigned unique ID, routed to Warden queue.
Requirement Mapping: FR-001 [Priority: High]	Postconditions (Failure): Submission rejected; no ticket record created.

3. Main Success Scenario

- Resident selects 'Log New Ticket' from the app home screen. System displays ticket form with fields for room number tag, category, and description.
- Resident confirms or edits the pre-filled room tag (auto-detected from profile).
- Resident selects an issue category (Plumbing, Electrical, Carpentry) and severity level (Low/Medium/High).
- Resident optionally attaches a photo of the maintenance issue (Attach Photo Evidence extension).
- Resident submits the ticket.
- System validates that all required fields (room tag, category, severity) are present.
- System generates a unique ticket ID, timestamps submission, and routes ticket to Maintenance Warden queue.
- System displays confirmation screen with ticket ID and estimated response timeline. Use case ends.

4. Alternate Flow — 6a: Incomplete Submission

- 6a. Triggered at step 6 if a required field (e.g., room number tag) is missing or invalid.
 - 6a1. System rejects submission and highlights missing field(s) in red.
 - 6a2. System displays error prompt: 'Please complete all required fields before submitting.'
 - 6a3. Resident inputs missing data and resubmits. Flow resumes at step 6 once validation passes.